VBLS — α sensitivity: non-monotonic $t_B(\alpha)$ with optimum $\alpha^* \approx 0.135$ yr^{-1}

Active regime ($\alpha < \alpha^*$): $\alpha \uparrow \Rightarrow \eta_{abs} \downarrow \Rightarrow F_{diss} \downarrow \Rightarrow \Sigma \downarrow \Rightarrow t_B \uparrow$ | **Collapse regime ($\alpha > \alpha^*$):** η_{abs} **collapses,** $E \to E^*$ **early** $\Rightarrow t_B \downarrow$







